Subclinical endothelial dysfunction and low-grade inflammation play roles in the development of erectile dysfunction in young men with low risk of ...

The purpose of this study is to investigate the possible underlying pathogenesis of erectile dysfunction(ED) in young men with low risk of coronary heart disease and no well-known aetiology. To conduct this study, 122 patients with ED under the age of 40 were enrolled, along with 33 age-matched normal control subjects. The patients with ED had significantly higher levels of systolic blood pressure (SBP), total cholesterol and triglyceride, high sensitivity C-reactive protein (hs-CRP), greater carotid intima-media thickness (CIMT) and Framingham risk score (FRS) than the control group, though all of these values were within the respective normal range. Further, the brachial artery flow- mediated vasodilation (FMD) values were significantly lower in ED patients and correlated positively with the severity of ED ($r = 0.714$, $p < 0.001$). When these significant factors were studied in the multivariate logistic regression model, FMD, SBP, hs-CRP and FRS remained the statistical significance. The receiver-operating characteristic (ROC) analysis demonstrated that FMD had a high ability to predict ED in young male with low FRS [area under the curve (AUC) 0.921, $p < 0.001$]. The cutoff value of FMD <10.25% had sensitivity of 82.8% and specificity of 100% for diagnosis of ED. FRS and hs- CRP were also proven to be predictors of ED (AUC 0.812, $p < 0.001$; AUC 0.645, $p = 0.011$, respectively). The results of this study validated that subclinical endothelial dysfunction and low-grade inflammation may be the underlying pathogenesis of ED with no well-known aetiology. Young patients complaining of ED should be screened for cardiovascular risk factors and possible subclinical atherosclerosis. Measurement of FMD, hs-CRP and FRS can improve our ability to predict and treat ED, as well as subclinical cardiovascular disease early for young male.